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Research Topic: Agricultural Drought Onset Across Romania: Evaluating Machine-Learning Early Warning at Multiple Lead Times

Predicting Agricultural Drought Onset Across Romania: Evaluating Machine-Learning Early Warning at Multiple Lead Times

Abstract

Agricultural drought represents a significant challenge for Romanian agriculture, creating a need for reliable early-warning systems that can identify drought conditions before their onset. Existing Romanian research has described the spatial and temporal variability of drought, while recent studies have also investigated machine-learning forecasting in individual regions such as Dobrogea. Nevertheless, the predictability of drought onset at specific lead times across the entire country remains insufficiently explored. This study proposes to investigate how accurately agricultural drought onset can be predicted by machine-learning models at lead times of 1, 2, 3 and 6 months, while examining how predictive skill varies geographically. Monthly ERA5-Drought and ERA5-Land reanalysis data from 1950–2025 will be used to construct SPEI-based drought-onset labels and climatic predictors including precipitation, 2-m temperature, soil moisture, evaporation, humidity, wind speed, and previous SPEI. A Logistic Regression baseline will be compared with Random Forest, XGBoost and LSTM models using temporally separated training, validation and test periods to prevent future information from entering earlier predictions. Model performance, geographic variation and predictor importance will be analyzed to determine the practical limits of ML-based drought early warning across Romania and identify which climatic variables provide useful predictive information at different lead times.

Introduction

Drought is a natural disaster that significantly impacts agriculture and its development. Because agriculture is a fundamental component of food production in Romania, the adverse effects of agricultural drought pose a serious challenge. Consequently, early prediction of the indicators and onset of a drought episode is essential.

However, this drought condition has not been uniform across the country, as Romania features a highly varied and heterogeneous topography. This spatial diversity introduces a substantial challenge for country-level drought prediction.

To address this forecasting difficulty across diverse regions, machine learning models offer an effective approach due to their ability to capture complex, non-linear climatic relationships. However, since drought is not an event of fixed duration but rather a variable one, forecasting it requires considering multiple time horizons. This study aims to investigate how early agricultural drought onset can be predicted across Romania and how predictive skill and relevant climatic predictors vary geographically.

Research Problem:

Drought represents an important climatic hazard in Romania, affecting water availability and agriculture. Extreme events occur most commonly over a period of 2 to 3 months, as shown in previous studies examining the Standardized Precipitation Index [1]. Ioniță et al. [2] found regional

differences, as well as a higher frequency of drought events in the southern and eastern parts of the country, suggesting that drought is not spatially uniform across Romania.

Knowing where drought has historically occurred is useful, but agricultural planning also requires knowing whether drought can be anticipated before it begins by being able to predict in advance over a period of 1 to 6 months. Traditional drought monitoring commonly relies on indices such as SPI or SPEI; nonetheless, recent Machine Learning research suggests nonlinear relationships between climatic variables can be exploited for drought prediction [3].

Recent work has already applied machine learning to drought forecasting in Dobrogea [4]. Dobrogea represents one of the south-eastern regions of Romania, being known for its generous portion of land devoted to agriculture. Agriculture is a key factor for the community, and drought episodes being more frequent in the last decades poses a significant challenge. The data gathered from 6 stations, at multiple time scales, have been explored with machine-learning models, trying to predict relevant drought indices such as SPI and SPEI. However, forecasting an index for one region is different from asking whether the transition into drought can be anticipated across an entire country. This study aims to further explore the onset of drought rather than general drought-index forecasting, as well as spatial variation across Romania and explicit forecast lead time.

Research questions

Main research question: How accurately can machine-learning models predict the onset of agricultural drought across various regions of Romania for a period of 1,2,3,6 months ahead using publicly available data?
Research question 1: How does predictive performance vary geographically across Romania?
Research question 2: Which climatic variables contribute most strongly to drought-onset prediction, and does their importance change with prediction horizon?

Hypothesis

1. Machine learning models will achieve a higher predictive skill for shorter horizons (1-2 month periods) compared to performance for longer forecast horizons, which will decline.
2. Models incorporating multiple climatic and land-surface predictors will achieve higher predictive skill than models based on precipitation alone.
3. Predictive skills will vary geographically because drought characteristics themselves vary spatially across Romania.

Literature Review

Standardized Precipitation Index (SPI) and Standardized Precipitation Evapotranspiration Index (SPEI) are widely used drought indices. The Standardized Precipitation Index (SPI) was developed by McKee et al. [5] to identify and track drought conditions. Existing research demonstrates substantial temporal and spatial heterogeneity. The findings of Ioniță et al. [2] pose a relevant example which highlighted the major influence of the Carpathians across the land of Romania, as well as the increased vulnerability for prolonged drought in the southern and eastern regions. This is supported by Bădăluță et al. [6] which sustained the fact that regions experience a different vulnerability to drought, Moldova, Oltenia and Banat being especially affected by drought episodes. Previous research [1]

shows that drought spells lasting 2 or 3 months are the most frequent, summing up >90% of the total drought cases as a country average. Therefore, all these findings highlight the significant heterogeneity of Romanian landform, justifying the evaluation of a spatial forecast, rather than a national aggregate.

Internationally, researchers are turning to machine learning to better predict droughts. For instance, a recent global study applied a LightGBM model to ERA5 climate data to forecast sudden flash droughts up to two weeks in advance. The results reported by Xu et al. [7] showed improved flash-drought prediction relative to the comparison model. More broadly, Oyarzabal et al. [8] found that most commonly the SPEI and SPI variables are used as predictors, but the precipitation as well; regression is the most commonly used method, at about 77% of the papers published utilizing. Beyond model choice, previous research also indicates that the selection and combination of predictor variables can influence drought-prediction performance. Nikdad et al. [9] combine ground observations, reanalysis and remote-sensing datasets to evaluate feature-selection techniques for agricultural drought prediction. Merging datasets improved the accuracy of datasets compared to in situ measurements. Combined, these studies indicate that drought-prediction performance depends not only on the selected machine-learning algorithm but also on the climatic information provided to it. This motivates comparing multiple model families and predictor configurations in the proposed study.

The most relevant and recent work in the area of the proposed research is Șerban et al. [4] which analyzes the drought in Dobrogea, south-eastern region of Romania, utilizing machine-learning models to predict the SPEI. They utilized data from 6 in situ meteorological stations, as well as a satellite-based dataset to forecast SPEI at multiple timescales (1-24 months). The models which performed best are represented by SARIMA and Support Vector Machines, providing both local and regional perspectives on drought variability. Nevertheless, Șerban only focuses on a restricted geographical area and predicts indices rather than specifically the transition into drought. This highlights that it does not directly address the question of how predictive skill changes at explicit 1/2/3/6-month lead times across Romania. Therefore, it is not constructed as a nationwide comparison of geographic predictability and variable importance.

METHODOLOGY

4.1 Study Area and Data

The study area is represented by the entire territory of Romania, and the temporal resolution will be monthly. The study will focus on the 1950-2025 time period, analyzing data from two datasets: ERA5-Drought [10], which provides SPEI used to construct the drought-onset target, and ERA5-Land [11], which provides meteorological and land-surface predictors. These two datasets are derived from the publicly available Copernicus/European Centre for Medium-Range Forecasts reanalysis products. The candidate predictors used as model inputs include: total precipitation, 2-m temperature, soil moisture, evaporation, dew-point temperature/relative humidity, and wind speed.

4.2 Drought-Onset Definition

Compared to other natural disasters, a drought does not happen suddenly; it is a slow-developing phenomenon. Because of this gradual shift, identifying the exact moment a drought starts can be rather complex. Therefore, the target variable will be defined using SPEI-3 [12], as it represents a

widely used drought index, combining the effects of precipitation and atmospheric evaporative demand.

$$D_t = \begin{cases} 1, & SPEI3_t \leq -1 \\ 0, & SPEI3_t > -1 \end{cases}$$

The target is not drought occurrence, but drought onset. Onset:

$$Onset_t = 1 \quad \text{if} \quad SPEI3_{t-1} > -1 \quad \text{and} \quad SPEI3_t \leq -1$$

An onset event is a transition from non-drought conditions during the previous month into drought conditions during the current month. This prevents an ongoing drought from being repeatedly classified as a new onset event.

4.3 Predictor Variables and Feature Engineering

Data will be divided into monthly grid observations, which will include the following candidate predictors: precipitation (P_t), 2-m temperature (T_t), soil moisture (SM_t), evaporation/potential evaporation (E_t), humidity (RH_t), wind speed (W_t), previous SPEI ($SPEI_t$). Moreover, previous monthly observations will be considered as lagged features, enabling the models to factor in the development of climatic conditions preceding a drought. Separate prediction targets will be made for drought onset 1, 2, 3, and 6 months ahead, utilizing only information available at the time when each prediction would be made.

4.4 Machine-Learning Models

4 The proposed models for this agricultural drought onset prediction and analysis are: Logistic Regression, Random Forest, XGBoost, and Long Short-Term Memory (LSTM). Logistic Regression represents the statistical/simple baseline, as a starting point. Random Forest and XGBoost will capture 4 nonlinear interactions between climatic variables. Long Short-Term Memory (LSTM) will model temporal dependencies across sequences, as it works with monthly sequences in order to learn temporal patterns from sequential observations across drought onset, justifying the inclusion of a deep learning sequence-based model.

4.5 Experimental Design and Validation

The dataset will be divided into subsets in order to better predict and validate the findings. The 1950-2025 study period will be split: Earlier Years, Training (1950-2005), Validation (2006-2015), Later Years, and Test (2016-2025). The reason for this split is to prevent future observations from being used to predict the past. Preprocessing, feature scaling and hyperparameter tuning will be utilized only for working with the training/validation data, whereas the test period will remain unseen until the final evaluation.

Predictions will initially be evaluated spatially across the Romanian grid rather than presuming equivalent predictive performance throughout the country. Performance can be aggregated into major geographic regions for interpretation. This analysis will identify whether certain areas of Romania permit earlier or more reliable drought-onset warning than others.

Data analysis plan

The results provided by the machine-learning models need to be evaluated and analyzed in order to highlight key information. Because drought onset is expected to occur less frequently than non-onset conditions, overall classification accuracy alone could be misleading. Therefore, utilizing relevant evaluation metrics is a key factor in the data analysis plan: Precision-Recall Area Under the Curve (PR-AUC), F1-score, precision, recall, Receiver Operating Characteristic Area Under the Curve (ROC AUC). Recall is relevant for early warning, because it measures the proportion of actual drought-onset events correctly detected.

To best evaluate outcomes, a table will be created which will highlight the results of every model at the specific time period (1, 2, 3 and 6 months). The performance decay between 1 and 6 months will test the first hypothesis. To test the second hypothesis, an ablation experiment will compare precipitation-only, precipitation-temperature, and full-predictor configurations. The performance metrics will be calculated geographically as well, in order to create a spatial map of prediction performance across Romania, rather than only at national level. Performance metrics will be calculated spatially and aggregated by predefined geographic regions and compared across the four forecast horizons. This analysis will investigate whether drought onset can be predicted more reliably or further in advance in some parts of Romania than in others.

For the Random Forest and XGBoost models, permutation feature importance will be used to evaluate the contribution of individual predictor variables. The values of each predictor will be shuffled and the resulting decrease in model performance will be measured. A larger decrease will indicate that the predictor contains more useful information for drought-onset prediction.

Timeline

Period	Research activity
Weeks 1-2	Data acquisition, Romanian spatial extraction and preprocessing
Weeks 3-4	Drought-onset labeling and feature engineering
Weeks 5-6	Baseline, Random Forest and XGBoost training
Weeks 7-8	LSTM training and multi-horizon evaluation
Weeks 9-10	Geographic analysis, feature importance and robustness testing
Weeks 11-12	Interpretation, visualization and final research paper

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